# Paper II — draft protocols and preparation supplement

DRAFT — NOT SUBMISSION READY: main study sealed; delegated mechanical adjudication; A/B/C/E AI-assisted review imported as a provisional sensitivity layer; investigator sign-off, human adjudication and human validation (Section D) pending

Table S11 gives analytic precision planning at assumed proportions. These values are not observed outcomes and do not replace design-based intervals.

*Table S11. Wilson planning intervals (unweighted binomial reference).*

| Planned n | Assumed p | 95% lower | 95% upper | Half-width |
|---|---|---|---|---|
| 100 | 0.2 | 0.133366933 | 0.288829166 | 0.077731116 |
| 100 | 0.3 | 0.218948853 | 0.395848546 | 0.088449847 |
| 100 | 0.5 | 0.403831530 | 0.596168470 | 0.096168470 |
| 100 | 0.7 | 0.604151454 | 0.781051147 | 0.088449847 |
| 100 | 0.8 | 0.711170834 | 0.866633067 | 0.077731116 |

*Table S3. Main-cohort paper dispositions (delegated; human adjudication pending).*

| Disposition | Papers |
|---|---|
| attempted failure | 10 |
| not attempted all inputs leak target | 10 |
| not attempted external reference resource required not served | 28 |
| not attempted gate failed delegated | 24 |
| not attempted no input retained | 1 |
| not attempted required inputs not in public listing | 16 |
| not attempted specification not obtained | 7 |
| not attempted target value not located verbatim | 4 |

*Table S4. Run-level L1–L5 states across 30 sealed slots.*

| Level | State | Runs |
|---|---|---|
| L1 specification extracted | no | 20 |
| L1 specification extracted | yes | 10 |
| L2 implementation created | not reached | 1 |
| L2 implementation created | yes | 29 |
| L3 execution successful | not established | 26 |
| L3 execution successful | not reached | 1 |
| L3 execution successful | yes | 3 |
| L4 numerical target reproduced | no | 2 |
| L4 numerical target reproduced | not established | 27 |
| L4 numerical target reproduced | yes | 1 |
| L5 conclusion preserved | not assessable | 29 |
| L5 conclusion preserved | pending human adjudication | 1 |

*Table S5. Failure taxonomy codes by run and by paper.*

| Code | Runs | Papers |
|---|---|---|
| F01 | 5 | 2 |
| F03 | 2 | 1 |
| F13 | 19 | 9 |
| F15 | 3 | 3 |
| F20 | 2 | 2 |

*Table S6. Descriptive original-code reveal ledger (post-seal, not executed).*

| Paper | Assessment | Dimension | Completeness | Description |
|---|---|---|---|---|
| PMID:33072212 | missing | | not available | |
| PMID:34495251 | missing | | not available | |
| PMID:34731224 | missing | | not available | |
| PMID:39830509 | missing | | not available | |
| PMID:39895948 | missing | | not available | |
| PMID:40956867 | described | normalization | not reported | not located in retained repository |
| PMID:40956867 | described | filtering | not reported | not located in retained repository |
| PMID:40956867 | described | missing data handling | not reported | not located in retained repository |
| PMID:40956867 | described | initialization | partial | simulation initialises each deme with msprime PopulationConfiguration(initial size=n e); n e defaults to 1 |
| PMID:40956867 | described | random seeds | not reported | sample sizes drawn with np.random.binomial without a seed argument in the retained sim.py; no fixed seed located |
| PMID:40956867 | described | stopping criteria | not reported | not located for the simulation study; optimizer stopping lives in the estimation code and was not traced to the scenario |
| PMID:40956867 | described | defaults | explicit | simulate genotypes w admixture defaults: chrom length=1, mu=1e-3, n e=1, target n snps=1000, n print=50, admixture props=[0], time of adm=[1], replic=0, dump=False |
| PMID:40956867 | described | preprocessing order | not reported | not located in retained repository |
| PMID:40956867 | described | exclusions | not reported | not located in retained repository |
| PMID:40956867 | described | constants and thresholds | partial | setup graph defaults barrier w=0.1, corridor w=0.5; admixture |

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| | | | | |
|---|---|---|---|---|
| | | | | proportion enters as MassMigration proportion=admixture props[i] |

| | | | | |
|---|---|---|---|---|
| PMID:40956867 | described | boundaries | partial | setup graph grid defaults n rows=8, n columns=12; scenario grids for the published simulation not located |
| PMID:40956867 | described | software versions | explicit | requirements.txt pins msprime==1.0.0, numpy==1.18.5, scipy==1.5.0, scikit-learn==0.23.1, networkx==2.4.0 |
| PMID:40956867 | described | manual transformations | not available | scenario-driving scripts for the dense-sampling / c=0.5 scenario were not located in either retained archive |
| PMID:41091855 | described | normalization | explicit | loads pre-computed LDSC output (LDSC ADGWAX QTRAIT.RData); no normalisation step in the deposited scripts |
| PMID:41091855 | described | filtering | explicit | none in the deposited scripts |
| PMID:41091855 | described | missing data handling | not reported | not addressed in the deposited scripts |
| PMID:41091855 | described | initialization | partial | GenomicSEM usermodel defaults; no explicit start values |
| PMID:41091855 | described | random seeds | not reported | no set.seed in either script |
| PMID:41091855 | described | stopping criteria | not reported | GenomicSEM/ lavaan optimizer defaults; not set in scripts |
| PMID:41091855 | described | defaults | explicit | QTrait(mresid=0.25, mresidthreshold=0.10, lsrmr=0.25, lsrmrthreshold=0.10, save.plots=TRUE, stdout=FALSE) |

| | | | | |
|---|---|---|---|---|
| PMID:41091855 | described | preprocessing order | explicit | indicators = first 3 columns of S; traits = remaining columns; common-factor model then common-pathway and independent-pathway models |
| PMID:41091855 | described | exclusions | explicit | none in scripts |
| PMID:41091855 | described | constants and thresholds | explicit | estimation="DWLS"; CFM std.lv=T, imp cov=F; CPM/IPM std.lv=F, imp cov as coded; residual thresholds as in defaults |
| PMID:41091855 | described | boundaries | not reported | not applicable in scripts |
| PMID:41091855 | described | software versions | partial | header states R (>= 4.0); GenomicSEM version not pinned |
| PMID:41091855 | described | manual transformations | explicit | results written to CSV; no manual step described |
| PMID:41472739 | missing | | not available | |
| PMID:41563087 | missing | | not available | |
| PMID:41677189 | described | normalization | explicit | DESeq2 DESeq() default size-factor normalisation |
| PMID:41677189 | described | filtering | explicit | genes with rowSums(counts) > number of samples retained in the interactive path; report scripts use rowSums(counts) > 10 |
| PMID:41677189 | described | missing data handling | explicit | results rows with NA removed by na.omit() before thresholding |
| PMID:41677189 | described | initialization | partial | DESeq2 defaults |
| PMID:41677189 | described | random seeds | not reported | none set |
| PMID:41677189 | described | stopping criteria | partial | DESeq2 defaults |

| | | | | |
|---|---|---|---|---|
| PMID:41677189 | described | defaults | partial | UI defaults adjusted p < 0.05 and \|log2FC\| > 1; design columns, contrast column and levels are user-selected at run time and are not recorded in the repository |
| PMID:41677189 | described | preprocessing order | explicit | DESeqDataSetFromMatrix -> low-count filter -> DESeq -> results(contrast) -> na.omit -> threshold |
| PMID:41677189 | described | exclusions | explicit | low-count genes only |
| PMID:41677189 | described | constants and thresholds | partial | padj cutoff and log2FC cutoff are interactive inputs; defaults 0.05 and 1 |
| PMID:41677189 | described | boundaries | explicit | slider ranges 0-0.2 (padj) and 0-3 (log2FC) |
| PMID:41677189 | described | software versions | not reported | no lockfile or sessionInfo in the retained archive |
| PMID:41677189 | described | manual transformations | ambiguous | count matrix and metadata uploaded interactively; the exact uploaded files and selections used for the published counts are not recorded |

*Table S7. Paper-level successful-slot distribution among 10 attempted papers: frozen mechanical versus verification-adjusted (AI-assisted review, provisional).*

| Successful slots | Frozen papers | Adjusted papers |
|---|---|---|
| 0/3 | 9 | 9 |
| 1/3 | 1 | 0 |
| 2/3 | 0 | 1 |
| 3/3 | 0 | 0 |

*Table S8. 100-paper funnel by state: frozen record, verification-adjusted state and unresolved envelope (no reclassified paper was executed).*

| State | Frozen | Adjusted | Envelope |
|---|---|---|---|
| all inputs leak target | 10 | 1 | 1 |
| external reference resource required not served | 28 | 28 | 28 |
| gate failed delegated | 24 | 25 | 24 |
| no input retained | 1 | 1 | 1 |
| required inputs not in public listing | 16 | 17 | 16 |
| slots completed | 10 | 10 | 10 |
| specifiable runnable | 0 | 5 | 0 |
| specification not obtained | 7 | 7 | 7 |
| target value not located verbatim | 4 | 4 | 4 |
| unresolved | 0 | 2 | 9 |

*Table S9. Descriptive reveal completeness corrections proposed by the E review (papers with a code route only; descriptive, not executed).*

| Paper | Dimension | Frozen | Adjusted |
|---|---|---|---|
| PMID:41091855 | exclusions | explicit | partial |
| PMID:41091855 | filtering | explicit | partial |
| PMID:41091855 | manual transformations | explicit | not reported |
| PMID:41091855 | normalization | explicit | partial |
| PMID:41677189 | exclusions | explicit | partial |

*Table S10. Human validation (Section D): PENDING — no participant, case or outcome data exist.*

| Item | Value |
|---|---|
| Cases completed | 0 (pending external human validation) |

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*Table S1. Conceptual mapping of reproducibility requirements to framework stages. ACCESS is refined into resource accessibility, input-state identifiability and historical-state retrievability; the mapping is conceptual, not an empirical result.*

| Requirement | Stage | Purpose |
|---|---|---|
| repository/accession | ACCESS | identify data source |
| release/version | ACCESS | identify exact dataset state |
| retrieval date/time | ACCESS | anchor dynamic resource state in time |
| exact query/API parameters | ACCESS | reconstruct query-generated input |
| file list/schema | ACCESS | fix input composition/structure |
| SHA-256/content hash | ACCESS / REPRODUCE | verify byte-level identity |
| transformation log | RECONSTRUCT / REPRODUCE | trace derivation from raw to analytical input |
| license/redistribution status | ACCESS | explain why exact snapshot can or cannot be redistributed |
| code availability | ACCESS (artifact access) | locate the original implementation; artifact reproducibility only, deliberately excluded from the Paper II blind intervention and used solely as a descriptive characteristic and post-seal reveal resource |
| software/version reporting | ACCESS / EXECUTE | identify the executable environment |
| container/environment | EXECUTE | reinstate the execution environment |
| detailed Methods | RECONSTRUCT | specify the procedure well enough to implement |
| parameters/defaults | RECONSTRUCT | fix analytical choices that change results |
| random seed | EXECUTE / REPRODUCE | control stochastic variation |
| resource requirements | EXECUTE | state compute, memory and time needed |
| independent reconstruction | RECONSTRUCT → REPRODUCE | test the description itself |
| robustness testing | ROBUST | vary reasonable choices (Paper III; excluded here) |

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*Table S2. ACCESS-stage input-state fields as recorded in the frozen main-study records (descriptive; fields not recorded are reported as such; no dataset drift was measured).*

| Item | Value as recorded | Source |
|---|---|---|
| repository/accession recorded | 100/100 papers | delegated specification listing (100 papers) |
| repository kinds in listings | bioproject 10; dataverse 4; dryad 13; figshare 25; geo 70; osf 2; sra study 5; zenodo 19 | delegated specification listing (100 papers) |
| file list recorded in listing | 89/100 papers | delegated specification listing (100 papers) |
| external public reference resources named by specifier | 119 entries in 51 papers | delegated specification listing (100 papers) |
| reference version/release recorded by the specifier from publication text | 100/119 entries | delegated specification listing (100 papers) |
| reference version/release not specified in publication text | 19/119 entries (11 papers with at least one) | delegated specification listing (100 papers) |
| not run because an external public reference resource was required | 28/100 papers (frozen barrier state, unchanged) | paper status.json |
| served inputs with SHA-256 recorded | 21/21 files | frozen target packages and retrieval ledger (10 attempted papers) |
| served inputs with retrieval timestamp recorded | 21/21 files | frozen target packages and retrieval ledger (10 attempted papers) |
| served inputs with repository-announced checksum | 7/21 files; 7 matched | frozen target packages and retrieval ledger (10 attempted papers) |
| repository version/release identifier of served inputs | not recorded (live service snapshots) | frozen target packages and retrieval ledger (10 attempted papers) |
| query/API parameters | anonymous GET of listed files; no query-generated input | frozen target packages and retrieval ledger (10 attempted papers) |
| schema/release identifier | not recorded | frozen target packages and retrieval ledger (10 attempted papers) |
| historical version check | not assessed | frozen target packages and retrieval ledger (10 attempted papers) |

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# Agent reconstruction protocol

**DRAFT — NOT FROZEN. No authorized main-run configuration exists yet.**

## Proposed pilot envelope

Per slot: four vCPU, 16 GiB RAM, 100 GiB scratch, no GPU, 120 active wall minutes, 250,000 provider-accounted input+output tokens, and 1,000 tool calls. These are planning ceilings, not measured usage or demonstrated available platform settings. Pilot must verify enforcement, provider accounting, total-elapsed cap, outage clock policy, maximum monetary cost and availability of the same model/version. An unverifiable model version is recorded as unknown and limits the claim. No main run starts with null budget/stop rules in the harness contract.

Use three fixed slots, same target, package hash, prompt, permissions, dependency policy, Internet access boundary and resource tier. If a second tier is essential, assign it prospectively by paper inputs and use it identically within that paper. The preparation audit agents are not experiment slots.

## Solver instructions to freeze verbatim after pilot

1. Inspect permitted publication and target; record observable ambiguities with source locations, not private reasoning traces.

2. Produce an independent implementation faithful to the single reported method. Do not consult the original implementation, other runs or forbidden tools.

3. Install only broker-approved general dependencies. Record exact versions.

4. Execute and debug code within the fixed budget. Fix coding/environment errors; do not vary analytical specifications, models, seeds or defaults to improve target agreement. Record each attempt and implementation revision.

5. Generate the target once the declared procedure works. Use only the frozen comparison criterion. Record absolute/component errors and the target-linked conclusion separately.

6. Stop at completed assessment, verified unavoidable barrier, budget limit, suspected contamination or controller stop. Do not silently restart.

7. Submit code, permitted outputs, execution records, assumptions, ambiguities, failure evidence and structured outcomes. Missing usage is unknown, never zero.

## Required records

Run IDs and paper IDs; model/version/mode and exposed settings; protocol/prompt/package/context/tool-registry/image hashes; exact UTC start/stop; stop reason; L1–L5; failure and attribution evidence; attempts/revisions/dependencies; elapsed time, tokens and costs; access events and contamination status.

Implementation is observed executable code, not a persuasive report. Execution requires saved stdout/stderr/status/output artifacts. Success requires an independently verifiable comparison; conclusion agreement does not substitute for numerical agreement. An independent blinded adjudicator resolves ambiguous scores before blind freeze, without inspecting original author code.

# Protocol amendment AMEND-2026-09-24-03: deposited-data eligibility criterion

**Status: author-instructed amendment, hash-frozen before any candidate is screened under it. Applies to every candidate selected after the freeze.**

## Basis

The provisional pilot (`DEV-2026-09-24-02`) reached no reconstruction attempt: none of the seven provisional papers had study inputs that a third party could obtain lawfully and immediately. On 2026-09-24 the author instructed that the main analysis of publication-grounded reconstruction be restricted to papers whose deposited analysis data are lawfully and immediately obtainable by a third party, that every excluded paper be retained with its exclusion reason, and that the seven provisional papers be moved to a separate accessibility layer without any reconstruction success or failure rate.

## Amended eligibility (replaces the G3-as-outcome rule for the main sample)

A paper is eligible for the main reconstruction sample only if, in addition to lawfully retained article text, primary G1 = yes and G2 != no, all of the following hold at screening time and are recorded with retained evidence:

1. **Deposited analysis data are named in the article.** The article or its open supplement names a repository accession, persistent identifier or supplementary data file that holds the data consumed by the target computation (not only the outputs of that computation).

2. **Third-party lawful access is immediate.** The listing endpoint of the deposit answers an anonymous request with a machine-readable file list. No account approval, data-access committee, data-use agreement, author contact, institutional licence or payment may stand between a third party and the bytes. Free registration that is granted automatically is acceptable only if the file list itself is anonymous.

3. **The deposit is practically retrievable.** Every file needed for the target computation has an announced size, each file is at most `FILE_CAP = 2 GiB`, and the needed files total at most `TOTAL_CAP = 8 GiB`. Deposits that announce no sizes, or whose needed files exceed either cap, are excluded as practically unretrievable.

4. **The deposit is not outcome leakage.** Files that directly contain the target value (figure source tables, result tables) do not qualify as analysis data; they are withheld from every solver and held for post-reveal validation.

## Exclusion classes (retained, never silently dropped)

| class | meaning |
|---|---|
| `no lawful text retained` | article text not lawfully retained; unadjudicated, not negative |
| `no deposit named` | no accession, persistent identifier or data file detected in the retained text |
| `author request only` | data stated as available on request or from the corresponding author |
| `application required` | controlled-access repository or consortium (e.g. ADNI, dbGaP, EGA, UK Biobank) |
| `institutional licence` | access depends on an institutional contract or licence |
| `listing unavailable` | named route returned no anonymous machine-readable file list |
| `size unannounced` | route lists files without sizes |
| `above cap` | needed files exceed `FILE CAP` or `TOTAL CAP` |
| `deposited output only` | the only deposited files are the target outputs (outcome leakage) |
| `code only deposit` | the only named deposit is source code (blocked by the firewall) |
| `primary G1 no` / `primary G2 no` | fails the unchanged computational gates |

Exclusion under this amendment is a **sampling frame restriction**, not a G3 study outcome. The main-sample estimand therefore becomes reconstruction success among computational papers with immediately obtainable deposited analysis data, and every manuscript statement of the estimand must say so.

## Accessibility layer

Papers excluded for access reasons after G1 = yes and G2 != no are retained in `data/adjudication/accessibility_gate_failed_*.json` with class, route evidence and the deposit statement segment IDs. The seven provisional pilot papers are the first members of this layer (`ACCESSIBILITY_GATE_FAILED`). This layer is an audit sample of pre-reproducibility access barriers for an auxiliary analysis or a separate paper. **No reconstruction success or failure rate may be computed from it**, because no member reached a reconstruction attempt; the validation code rejects any rate field in the layer record.

## Sequence

1. Freeze this amendment (SHA-256 + RFC 3161 receipt) before any screening.

2. Screen candidates in the unchanged deterministic order per stratum, retaining every listing response and every exclusion reason.

3. Record primary G1–G5 for the first deposit-eligible candidate per stratum as a delegated assessment pending author verification.

4. Only then freeze pilot selection under the amended criterion.

# Protocol amendment AMEND-2026-09-25-04: pilot role, investigator verification and prospective target freeze

**Status: investigator-instructed amendment, hash-frozen before any candidate selected under AMEND-2026-09-24-03 is verified, frozen or reconstructed. Applies to every pilot and main-study record written after the freeze.**

## Basis

On 2026-09-25 the principal investigator instructed that (1) the seven candidates selected under AMEND-2026-09-24-03 are a prospective pilot / pipeline-validation set and not the main analysis sample; (2) G1–G5 verification and a prospective target specification must be complete for all seven before any blind reconstruction begins; (3) the verifier is the research team, so records must say "investigator verification" or "human adjudication", not "author verification", and original authors are not contacted for the primary reconstruction protocol; (4) accessibility and screening failures stay in the full-corpus attrition layer with machine-readable reasons; (5)–(6) the deposited-data and leakage rules of AMEND-2026-09-24-03 remain in force; (7) ten fields are frozen per pilot paper before reconstruction; (9) pilot results are used only to find procedural defects, never to raise the observed success rate; (12) no Paper III analysis is performed.

## 1. Terminology

`investigator_verification` replaces `author_verification` as the key and phrase for the research team's review of delegated (Devin) G1–G5 gates. Records written before this amendment keep their original key; readers accept both. "Author" is reserved for the original study authors and for the investigators' own authorship of this study. The primary reconstruction protocol never contacts original authors; author clarification is not an allowed resource.

## 2. Role of the seven AMEND-2026-09-24-03 candidates

The candidates selected under AMEND-2026-09-24-03 form the `PROSPECTIVE_PILOT` set: `main_sample_membership = excluded_from_primary_denominator`. Their outcomes validate the pipeline (target rules, tolerance rules, failure taxonomy, logging schema, resource ceilings) and are reported descriptively. They enter no primary reconstruction-rate numerator or denominator. The main cohort is drawn after the post-pilot protocol freeze from the full eligible population by the SAP's precision calculation and stratified random sampling (target N ≈ 100), with pilot papers removed from the sampling frame.

## 3. Additional exclusion class

| class | meaning |
|---|---|
| `deposited input mismatch` | the named deposit resolves anonymously but its listed samples/files are not the datasets the article reports analysing (identifier sets disjoint or incomplete); the target inputs are therefore not deposited under the named route |

The class is recorded with the article's own identifiers and the deposit's listed identifiers as evidence. It is a sampling-frame restriction like every AMEND-2026-09-24-03 class, not a reconstruction outcome, and the deterministic per-stratum chain resumes at the next rank.

## 4. Prospective freeze fields (per pilot and per main-study paper)

Before any slot is opened, the target manifest records, and the freeze hashes:

1. `exact_target` — the numerical value(s) exactly as published, with units and reporting precision;

2. `target_location` — section, table, figure or sentence, with the retained segment ID and verbatim quote;

3. `required_inputs` — repository, accession, file names, versions or run/sample identifiers, bytes and SHA-256 where retained;

4. `allowed_resources` — free software, public references and the package contents served to every slot;

5. `blocked_original_artifacts` — original code, repositories, notebooks, containers, processed result files and author clarification, all withheld;

6. `comparison_metric` — the TOLERANCE_RULES scorer applied;

7. `numerical_agreement_criterion` — the prospective numeric rule with reporting precision;

8. `conclusion_criterion` — the minimal target-linked claim and how agreement or disagreement with it is decided;

9. `resource_ceiling` — wall-clock, model-call, token and byte ceilings per slot;

10. `stopping_rule` — what ends a slot (report, ceiling, unresolved input, specification gap) and how each end state is labelled.

Known specification gaps (for example an undisclosed train/test split) are written into the manifest before the slot opens; a gap is a recorded `specification_insufficient` state, never a reason to substitute an easier target after the run.

## 5. Investigator verification record

For every pilot paper the investigator team records, gate by gate, `confirm` or `overturn` of the delegated value with a reason, after re-reading the cited retained segments and re-checking every quote verbatim. Overturns are kept alongside the delegated value. The record is validated against the retained text and the amended record before the target freeze may reference it.

## 6. Paper III boundary

No multiverse analysis, no deliberate alternative reasonable implementation, no robustness perturbation and no test of whether an undocumented implementation choice changes a conclusion is performed in Paper II. The original-code reveal after the blind freeze is descriptive only.

## Sequence

1. Freeze this amendment (SHA-256 + RFC 3161 receipt).

2. Complete deposit inspection, investigator G1–G5 verification and the ten freeze fields for all seven pilot candidates; resume a stratum's chain where a candidate is excluded.

3. Freeze the pilot target manifest (SHA-256 + RFC 3161), then run three independent blind slots per paper.

4. Pilot review; freeze the final protocol and SAP; select the main cohort.

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# Protocol amendment AMEND-2026-09-25-05: post-pilot final protocol freeze

**Status: written after the seven-paper prospective pilot was executed and sealed (blind outcome SHA-256 `485967ff8a1caf85c1d9b1bc102ada617219e3636ea22b7dcbbc5337bb81f3db`) and after the prespecified procedural review `PILOT-REVIEW-2026-09-25`, before any pilot value was adjudicated, before any original implementation was revealed, and before any main-study paper was selected. It freezes the reconstruction protocol, target rules, tolerance rules, failure taxonomy, logging schema and resource ceilings for the main study. Hashes of every frozen document are in `data/adjudication/protocol_freeze_20260925.json`, RFC 3161 timestamped under `data/raw/protocol-freeze-20260925/`.**

## Basis

The procedural review (data/adjudication/pilot_procedural_review_20260925.json) examined only harness behaviour: stop reasons, ceilings, journal completeness, report schema and value provenance. It computed no success or failure rate and compared no observed value with a publication. Its findings and the rule each receives are below. No target, tolerance or conclusion rule changes; nothing below can raise or lower the observed success rate of any pilot slot, because pilot slots are not re-run and remain outside the primary denominator.

## 1. PD-01 — resource ceilings made internally consistent

Sixteen of twenty-one pilot slots stopped at the cumulative provider-token ceiling (400,000 tokens) after a median of 23.5 tool calls and at most 139 s of a 14,400 s wall ceiling; no slot reached the 40-call ceiling and no slot ran a heavy computation. The token ceiling therefore silently overrode the tool-call and wall ceilings the protocol described as binding. The final ceiling sets the cumulative provider-token ceiling to **1,500,000 tokens per slot**, the value that makes the 40-call ceiling attainable at the 95th-percentile observed context size (31,844 prompt tokens × 40 = 1,273,760, rounded up to the next 500,000). Wall (14,400 s), step (3,600 s), tool-call (40), completion (8,192), memory (3 GiB), single CPU, no network and the 64 MiB artifact export cap are unchanged. The ceiling is a harness consistency correction and is applied to all main-study slots identically; it is never adjusted per paper.

## 2. PD-02 — failure codes are taxonomy identifiers

Reports carried free-text tokens (`NO_ANALYSIS_EXECUTED`, `RUN_TERMINATED_LENGTH_LIMIT`, `TARGET_NOT_OBSERVED`) in `failure_codes`. Final rule: `failure_codes` must be identifiers `F01`–`F20` or `F99` from FAILURE_TAXONOMY.md. Any other token is retained verbatim in the sealed report, mapped to `F99` for tabulation, and adjudicated as *unclassified*. No code is added to the taxonomy after this freeze.

## 3. PD-03 — final report without any execution

One report was submitted before any worker execution and asserted a harness termination that the journal does not show. Final rule: a final report with zero journaled executions is classified `no_execution_report`. The slot counts as attempted; L2 and L3 are `not_reached`; any `observed_value` is void; any procedural statement in a report is checked against the journal and never accepted from the report alone.

## 4. PD-04 — value provenance is required for L3 and L4

The served publication text displays the target value, so a value transcribed from the article is indistinguishable, in the report alone, from a computed one. Final rule: for L3 (`execution_successful`) and L4 (`numerical_target_reproduced`) the adjudicator must locate the observed value (to the reported precision) in a journaled execution output produced by journaled code that reads served inputs. If it cannot be located, L3 and L4 are `not_established` and the slot is recorded with `F15` if code exists or `no_execution_report` otherwise. The blind is a blind to the original implementation and its outputs, not to the publication; the manuscript states this and reports the provenance check explicitly.

## 5. Frozen documents

PROTOCOL.md, SAP.md, TOLERANCE_RULES.md, FAILURE_TAXONOMY.md, TARGET_SELECTION_RULES.md, INFORMATION_FIREWALL.md, REVEAL_PROTOCOL.md, this amendment and its predecessors, the logging schema (journal event kinds and the report schema in `src/paper2/controller.py` and `src/paper2/isolation.py`) and the final ceiling in `src/paper2/protocol_freeze.py`. Their SHA-256 values are listed in the freeze record. After the freeze, any change to these documents requires a new numbered amendment, and any change made after a main-study slot has opened is reported as a deviation.

## 6. Main study sampling frame

The eligible reconstruction population is every unique-PMID paper for which Stage A (deposit statement in lawfully retained text) and exhaustive Stage B (anonymous listing resolved, within caps, not implementation-only) record a verified open deposit route, minus (a) the seven `PROSPECTIVE_PILOT` papers, (b) the seven `ACCESSIBILITY_GATE_FAILED` papers, (c) every paper carrying a machine-readable exclusion class, and (d) every paper whose retained text was inspected for any earlier candidate review. The SAP allocation (one place per nonempty field, remaining places by proportional-allocation deficit with lexicographic ties, within-field order by SHA-256 of `paper2-main-v1|paper_id`) is executed once; $N_h$, $n_h$, $\pi_h$ and weights are saved before any paper's G1–G5 or target freeze begins. Each selected paper then receives the ten-field prospective freeze of AMEND-2026-09-25-04 §4 with the same G1–G5 delegated assessment and pending investigator verification; papers failing G1–G5, leakage review or deposited-input matching are recorded with their class and are not replaced (the primary estimate is reported on attempted papers with the attrition table; the SAP's finite-population bounds cover the unattempted).

## 7. Unchanged

Target selection hierarchy; tolerance rules; L1–L5 definitions and the primary endpoint (≥ 2 valid successes among 3 fixed slots per paper); three sequential single-VM slots; the information firewall; descriptive-only reveal; no Paper III analysis; no original-author contact; pilot outside the primary denominator.

DRAFT — NOT SUBMISSION READY: main study sealed; delegated mechanical adjudication; A/B/C/E AI-assisted review imported as a provisional sensitivity layer; investigator sign-off, human adjudication and human validation (Section D) pending

# Protocol amendment AMEND-2026-09-25-06: main-study execution layer

**Status: written after the final protocol freeze `PROTOCOL-FREEZE-2026-09-25` (SHA-256 `f3bdc79848672d06785081e339c8e6c26da26cae73b878d49687b933c2f4ef0f`) and after the main cohort `MAIN-COHORT-2026-09-25` (SHA-256 `c93708937c2ccf7f67d2e9c79b30db549caf6112ef7e5f82dd93e2becb286b4f`) was selected and timestamped, before any main-study paper was specified, before any main-study input was retrieved and before any main-study solver call. It records how the frozen protocol is executed for the cohort and one clerical correction. No target, tolerance, conclusion or failure rule changes.**

## 1. Clerical correction to TARGET_SELECTION_RULES.md

The document hashed by the final freeze (SHA-256 `fc879735acc5698093d1476db7e4d87ac4174807f5cbea0eb2068e91162a99da`) still carried the header line `**DRAFT — NOT FROZEN**` although AMEND-2026-09-25-05 froze it by hash together with every other protocol document. The header is replaced by a frozen-status line that cites this amendment. No rule text changes; the freeze record keeps the original hash, and the corrected hash is recorded in `data/adjudication/main-study-20260925/` at the first main-study freeze.

## 2. Delegated G1–G5 assessment and ten-field specification

For every cohort paper, in the order of `results/main_cohort_20260925.csv`, a delegated model pass (the same provider/model as the solver, deterministic settings, one request, journaled under `data/raw/main-study-20260925/ specifier-api/`) receives the segmented article text and the anonymous deposit file listing retained at Stage B. It returns G1–G5 with verbatim quotes, the exact target and its printed values, a blind description of the target, the target type, location, required deposit files, leakage-suspect files, external reference requirements, comparison metric, agreement criterion, specification gaps and rejected alternatives.

The harness verifies programmatically that every quote occurs verbatim in the cited segment or its immediate neighbours (segmentation is a harness artifact that can split sentences), that every printed target value occurs in the cited location quotes, that the blind description contains no printed value, that the target type has a complete comparison rule in TOLERANCE_RULES.md and that required files exist in the listing. A paper is `specifiable` only if no gate is `no`, the target value is located verbatim, at least one listed input is required and no external reference resource beyond the served material is required. Otherwise the paper is recorded with its machine-readable state (`gate_failed_delegated`, `target_value_not_located_verbatim`, `required_inputs_not_in_public_listing`, `external_reference_resource_required_not_served`, `specification_not_obtained`) and is **not** run. It stays in the cohort and in the denominator defined by the SAP for post-selection ineligibility; it is never replaced. Every delegated assessment is `investigator_verification: pending` and `human_adjudication: pending`.

## 3. Input acquisition and leakage exclusion

Only the files the specification names are retrieved, anonymously, from the public deposit route retained at Stage B, streamed to `/home/ubuntu/paper2_evidence/main-inputs-20260925/<PMID>/` with an immutable receipt (URL, identifier, UTC time, request conditions, bytes, SHA-256, announced checksum, completeness). Frozen caps: 2 GiB per file, 8 GiB per paper, retrieval stops when local free storage would fall below 12 GiB (recorded as `local_storage_exhausted`, not as an access verdict). A file named by the specifier as answer-bearing, or in which any printed target

value occurs as a whole numeric token, is excluded as `target_or_data_leakage_*` and never served. Result-like header columns are recorded for adjudication but do not by themselves exclude a file. A paper whose every required input is leakage-excluded is recorded `all_inputs_leak_target` and not run.

## 4. Per-paper prospective freeze

Before a paper's first slot its ten-field record (exact target, location with quotes, required inputs with SHA-256 and bytes, allowed resources, blocked original artifacts, comparison metric, agreement criterion, conclusion criterion, final resource ceiling, stopping rule) is written to `data/adjudication/main-study-20260925/freezes/<PMID>.json` and RFC 3161 timestamped. The solver instruction carries only the blind description.

## 5. Slots, order and quota stop

Exactly three sequential independent slots per specifiable paper, each in a fresh isolated worker under the final ceiling of AMEND-2026-09-25-05 (1,500,000 provider tokens; other limits unchanged), on one machine. Papers are processed strictly in cohort order. Execution ends when the investigator's provider quota is exhausted (three consecutive slots ending in a controller error before any tool call, or a provider stop during specification) or when the cohort is complete. Papers not reached, or reached without all three slots completed, are sealed as `not_run_resource_exhausted`; they remain in the cohort as unresolved and are never replaced or counted as failures. Completed slots are sealed before adjudication or any reveal.

## 6. What this amendment does not do

It does not change any frozen rule, does not re-run or re-score the pilot, does not compute any success or failure rate, does not contact original authors and does not perform any Paper III analysis.

## 7. Deposited code is withheld (DEV-2026-09-26-01)

On the first specifiable cohort paper the delegated specifier named deposited R scripts in `required_files` and the harness served them. Deposited scripts, notebooks and workflow files are original implementation artifacts under INFORMATION_FIREWALL.md and are never served; they are excluded as `original_implementation_artifact_withheld` by file name, the specifier is told not to list them, and a paper whose every required input is code is recorded `all_inputs_original_implementation` and not run. The two started slots are void (retained under `voided/`, in no denominator); the paper is re-frozen and run with three fresh slots. Details: `data/adjudication/main-study-20260925/ deviations/DEV-2026-09-26-01.json`.

## 8. OSF listings are resolved from the retained provider listings (DEV-2026-09-27-01)

On the 41st cohort paper the harness read an OSF route from its GUID record, which is a node object rather than a file list, and stopped with an exception before the specifier was called. The screening layer had already retained the per-provider file listings for every OSF route; the resolver now reads those sha256-verified bodies, skips folders, shows files as `<provider>:<name>` and serves them under the bare file name. No new listing request is made, no slot, instruction, ceiling or rule changes, and the manifest is superseded and retained. Details: `data/adjudication/main-study-20260925/deviations/ DEV-2026-09-27-01.json`.

# Protocol amendment AMEND-2026-09-25-07: primary adjudication, descriptive reveal and prespecified analysis

**Status: written after the main blind seal (`blind_outcome.json`, SHA-256 `350a0e7dcedbb74ccad2472c642318c87cc7d272f1b6c88186a1419dc858ea5a`, RFC 3161 timestamped) and before any original corpus-paper implementation was read. It records how the frozen rules (PROTOCOL-FREEZE-2026-09-25) are applied to the sealed outcomes. No target, tolerance, conclusion, failure or estimand rule changes. Human adjudication and investigator verification remain pending.**

## 1. Delegated primary adjudication (`src/paper2/main_adjudication.py`)

The sealed blind outcome is adjudicated mechanically from the frozen ten-field specification, the retained served inputs, the journaled code events and the journaled execution outputs of each slot. Original implementations are not opened. Rules applied:

- PD-04 provenance: a target component counts as observed only when it appears,

numerically identical, in the output of a successful execution of journaled code that reads the served inputs. Report-only values are `report_only` and set `human_review_flag` when they coexist with observed values.

- A slot with no journaled execution is `no_execution_report`; L2/L3 are

`not_reached` and no taxonomy code is assigned unless a protocol stop code applies.

- Stop reasons map to F13 (`wall_limit`, `tool_limit`, `token_reserve_limit`)

and F15 (`malformed_action_limit`). F20 is reserved for an executed run whose observed value does not reproduce the target. F17 marks contaminated slots as void, never as ordinary failure. Solver-reported reason text is kept verbatim and mapped to F01/F03/F20 only by the fixed patterns in the module; otherwise F99.

- Paper level: majority ($t=2$) primary; strict and permissive as sensitivities;

incomplete triples are never reduced to 0/3–3/3.

The first mechanical record (SHA-256 `8fab19e0…4664a`) assigned codes to no-execution slots, accepted approximate provenance matches and conflated L3 with L4. It was superseded before any human adjudication by the corrected record (SHA-256 `6fd42b89…5b28`); both records and both timestamp receipts are retained (`superseded/README.md`). Blind outcomes were not changed.

## 2. Descriptive reveal (`src/paper2/reveal_ledger.py`)

REVEAL_PROTOCOL.md was frozen by hash in AMEND-2026-09-25-05; its stale `DRAFT — NOT FROZEN` header is replaced with a frozen-status line and no rule text changes. Public original-code archives named in the retained article text are acquired after the seal (receipts with URL, bytes, SHA-256 and UTC time in `/home/ubuntu/paper2_evidence/main-reveal-20260925`). Paper-specific scripts that were part of a data deposit and withheld from the solver (`original_implementation_artifact_withheld` in the acquisition ledger) may be described from the withheld copy. Every quoted snippet is verified verbatim against the retained bytes; every source acquisition time is verified to postdate the seal unless it is a withheld deposit file. Papers whose publication names no code route receive a `missing` reveal assessment. Nothing is executed; no blind score is revisited.

DRAFT — NOT SUBMISSION READY: main study sealed; delegated mechanical adjudication; A/B/C/E AI-assisted review imported as a provisional sensitivity layer; investigator sign-off, human adjudication and human validation (Section D) pending

## 3. Prespecified analysis (`src/paper2/main_analysis.py`)

SAP.md is applied to the route-eligible population $|E|$ = 461 with the seven frozen strata. Non-attempted cohort papers are policy non-success under the recorded delegated states (`primary_policy`), and additionally treated as unresolved in a `verification_pending_envelope` because investigator verification of the gate states is pending. The conditional rate among attempted papers is secondary. Run-level L1–L5, stop reasons, slot states and failure codes are tabulated by run and by paper. No model, perturbation, multiverse or alternative-implementation analysis is performed.

# Commercial software policy

**DRAFT — NOT FROZEN**

Study-specific commercial software purchase equals zero. Do not use SAS, SPSS, Stata, MATLAB, proprietary modules, paid institutional entitlements or trials that cannot be replicated by a reader under the policy. Do not buy a licence.

A named commercial application is not automatic failure. Implement a sufficiently specified, software-independent method using free/open-source tools; record library versions and observable assumptions. Essential unreconstructable proprietary function = F02; required unavailable version = F03; underspecified procedure = the applicable specification code, not automatically F02.

Dependencies must be vetted as general-purpose tools, pinned with hashes and licences, and must not contain the paper's original implementation. The same dependency policy applies to every run. Agent/API costs, computing costs and journal APCs are distinct from software purchase; report actual usage or unknown. This draft does not authorize a new paid resource commitment.

# Failure taxonomy

**FROZEN 2026-09-25 by AMEND-2026-09-25-05. No code is added after the freeze.**

| Code | Category |
|---|---|
| F01 | input unavailable |
| F02 | commercial dependency |
| F03 | software or version unavailable |
| F04 | algorithm underspecified |
| F05 | parameter missing |
| F06 | preprocessing unspecified |
| F07 | data transformation ambiguous |
| F08 | software default unspecified |
| F09 | randomization or seed unspecified |
| F10 | numerical implementation ambiguous |
| F11 | hidden manual step suspected |
| F12 | dependency or environment failure |
| F13 | computational resource limit |
| F14 | publication internal inconsistency |
| F15 | agent reasoning or coding failure |
| F16 | target result insufficiently reported |
| F17 | firewall contamination |
| F18 | input identity or version ambiguous |
| F19 | access or download failure |
| F20 | runs but target not reproduced |
| F99 | other |

Record primary and optional secondary codes, earliest failed level, source location or execution evidence, observed event, hypothesized attribution and certainty separately. Allow uncertain, multiple and not_classifiable. For simultaneous unresolved causes do not force an arbitrary primary code.

A documented failed download is F19, not proof of F01. An incorrect agent program is not automatically F04. A difference from author code after reveal is descriptive evidence, not proof that the publication made reconstruction impossible. F11 remains suspected unless evidence establishes a hidden step. F17 invalidates the slot for clean inference; it is not an ordinary failed reconstruction.

Report `failure_codes` must be identifiers from this table. Any other token is retained verbatim, tabulated as F99 and adjudicated as unclassified. A final report with zero journaled executions is the slot state `no_execution_report` (L2 and L3 `not_reached`, observed value void), not a taxonomy code.

DRAFT — NOT SUBMISSION READY: main study sealed; delegated mechanical adjudication; A/B/C/E AI-assisted review imported as a provisional sensitivity layer; investigator sign-off, human adjudication and human validation (Section D) pending

# Blinded human validation

**NOT COMPLETED. This is a protocol and empty data-entry package, not human data.**

Purpose: assess the validity of the agent measurement process. It does not estimate human reconstructability in the corpus.

After blind freeze, sample approximately fifteen cases: five complete 3/3, five mixed (1/3 or 2/3), five complete 0/3. Within group use a frozen seeded hash order, retaining selection probabilities. If sparse, distribute unfilled places round-robin across remaining groups in fixed 3/3, mixed, 0/3 order, without exceeding group size. If fewer cases exist overall, report the reduced size; do not manufacture cases. Incomplete/contaminated triples are not silently assigned to an outcome stratum.

The coordinator alone sees selection labels. Give validators anonymous case IDs, the exact blind publication/input package, target and instructions, without agent code/logs/outcomes, original implementation or reveal findings. Packages cannot yet be assembled because no target packages or frozen outcome strata exist. The release build supplies the data-entry schema and these instructions.

Validator instructions: use the same allowed information, independently implement the stated method, record observable decisions and errors, and submit code, execution output, numeric comparison and target-linked conclusion. Record expertise, time limits, resources, any permitted assistance and deviations. Do not claim identical time/compute conditions if they are not actually comparable.

Before recruitment obtain the responsible institution's ethics determination for collecting human performance data, document actual consent arrangements and privacy/release rules. No exemption, consent, investigator identity or participant has been presumed. Real human participation must be supplied by the researchers.

Score with frozen rules using an adjudicator masked to agent outcome and sampling group. Record reviewer disagreements before consensus. Report a paired case table, discordant cases with evidence, and agreement proportion with its denominator; any binomial interval is descriptive conditional on the purposive outcome-balanced sample. Do not extrapolate it to all humans or all papers. Do not simulate responses. Empty `human_results.csv` remains NOT COMPLETED until actual submissions exist.

# Information firewall

**DRAFT — PARTIALLY QUALIFIED. Separate VMs alone are insufficient.**

`paper2.isolation` and `paper2.model_api` provide synthetic execution-isolation and API smoke qualifications only. `paper2.firewall` qualifies the custodian, broker, access telemetry, dependency closure, resource stop and freeze-before-reveal controls on a retained pilot article with adversarial negative cases (`results/firewall_qualification.json`, policy `firewall-qualification-2026-09-24`). Still unqualified: model-side retrieval tool restriction for provider-hosted tools, three-slot scheduling and budget comparability, and independent human adjudicator governance. None of these qualifications authorizes main runs. The provisional pilot (`paper2.pilot`, deviation `DEV-2026-09-24-02`) ran inside these qualified controls: one broker journal, one served package and one fresh worker per slot, manifests hashed and externally timestamped before the first solver call, and outcomes sealed and timestamped before any reveal-class request. The unqualified areas above still apply to it and are recorded as limitations of those runs. The current execution probe uses one CPU, 2 GiB RAM and 512 MiB work storage; it does not validate the larger proposed scientific-run envelope. A model alias without a provider-reported version remains explicitly version-unverified.

Allowed: publication text, methodological supplements including published scientific pseudocode, vetted public inputs, necessary general references and free/open-source documentation. Forbidden: original repository, source, notebook, workflow, container, alternative online reimplementations, and author material that reveals implementation rather than scientific specification.

Use separate custodian, controller/broker, solver and adjudicator roles. A custodian quarantines mixed archives and serves only reviewed exact members with archive-to-member provenance. Authors' implementation is not opened to prepare blind outcomes. Unknown/unchecked material is denied, even at an allowed domain. Preserve content hash, bytes, licence, acquisition conditions, reviewer and decision for every allowed or blocked artifact. Templates are in `schemas/`.

Each solver requires a new conversation, dedicated clean execution environment, no writable shared volumes, no cached browser or shell credentials, and no ancestor/sibling results. Shared read-only vetted software and paper inputs are permitted. Restrict BOTH execution networking and model-side web/Git/MCP/session tools. Audit injected rules, skills, knowledge, attachments, initial context and effective tool registry. A container network restriction does not demonstrate that model-side retrieval tools are restricted.

No parent debugging, package enrichment or policy widening during a run. Requests to the broker must be logged; allow only materials under the frozen policy that can be made equivalently available to all slots without outcome leakage. Record attempts, denials, bytes served, timestamps and immutable event hashes. Never record private chain-of-thought. Prior model-training exposure cannot be certified absent and is a limitation even with an enforced runtime boundary.

Qualification must demonstrate: forbidden canary artifact cannot be read by shell/browser/model-side tools; archive member and transitive dependency controls; no cross-run read/write; effective-context inventory; complete access telemetry; fixed resource stops; and frozen-outcome-before-reveal enforcement. These checks must themselves have evidence and adversarial negative cases. The preparation workflow that produced the protocols and candidate screening must not be relabelled a blinded reconstruction experiment; the firewall qualification run is a control test, not an outcome.

Contamination: stop, seal evidence, adjudicate independently. Suspected or unknown firewall status is unresolved for the clean primary analysis; confirmed contamination is excluded, retaining

its fixed slot as missing. No silent replacement or repeated attempts until success. The retained contaminated trace is not supplied to another solver.

# Paper II protocol

**FROZEN 2026-09-25 by AMEND-2026-09-25-05 after the seven-paper prospective pilot (`PILOT-FREEZE-2026-09-25`) was executed, sealed and procedurally reviewed. The frozen text hash and RFC 3161 timestamp are in `data/adjudication/protocol_freeze_20260925.json`. Investigator verification of delegated G1–G5 gates remains pending; pilot and accessibility-layer papers are excluded from the primary denominator. No main-study slot had opened at the freeze. Later changes require a numbered amendment.**

## Question and scope

Can publication-level scientific description support independent reconstruction of a computational research system? Define *publication-grounded independent computational reconstruction* operationally, with a specified agent, information boundary, resource budget, target and comparison rule. No claim of a new term or first agentic reconstruction study is made.

The framework is ACCESS → RECONSTRUCT → EXECUTE → REPRODUCE → ROBUST. Paper II measures reconstruction and its execution/numerical/conclusion consequences. ROBUST is future Paper III work and is excluded. The Wet/Dry mapping is a conceptual analogy, not a literal equivalence or an empirical result.

## Frame and eligibility

The source is the VOR-linked EPJ deposit, identified in the acquisition ledger. Preserve all rows. The automatically generated source audit distinguishes source records, unique PMIDs and overlapping field memberships. The requested distinct paper count is not met by the inspected deposit. No new papers are substituted. The author approved the unique-PMID inference frame while retaining all original rows. The verbatim decision is in `data/frame_decision.json`; the generated `results/frame_manifest.json` binds the decision, source and inference-frame hashes. This resolves the Paper II identity rule, not the historical distinct-paper claim, classification validity or prospective protocol freeze.

Paper identity is PMID; duplicate DOIs or titles trigger review, not automatic merging. Records are traceable by commit, file and one-based data-row position. Field labels describe recorded sampling provenance, not mutually exclusive scientific disciplines. Neither papers nor historical sample selection can be generalized to all PubMed or all science using unverified inclusion probabilities.

G1: yes/no/uncertain computationally testable, with evidence and reason. Include a central claim produced by a specified computational transformation, statistical estimation, simulation, optimization or predictive evaluation. Descriptive quantitative analysis is eligible when reproducing its central reported output requires applying a computational procedure to inputs. A purely verbal claim or restatement of a supplied number is ineligible. Missing inputs or inadequate reporting do not turn an otherwise computational paper into G1=no.

G2: yes/no/uncertain identifiable principal target under the frozen hierarchy. G3: public / obtainable_without_payment / restricted_but_potentially_accessible / unavailable / insufficiently_identified / not_applicable / unknown. G4: required function available under the commercial-software policy, with separate version and access reasons. G5: sufficient / insufficient / uncertain specification for an attempt. G6–G9 concern created implementation, execution, numerical agreement and preservation of the target-linked conclusion.

Store gates independently and derive sequential at-risk sets. Preserve unknown, unassessed and inaccessible categories. No gate is inferred from an EPJ regex flag. Ordinary free public registration

may be allowed if available to any reader; individual institutional permissions are restricted even when no fee is charged. Transient failed downloads remain unknown after a documented limited retry policy, not evidence that the underlying data do not exist.

Article-text obtainability is recorded separately from G1–G5 as a prospective access status: `lawful_text_retained` (identity-verified open PMC JATS, a licensed publisher/repository copy, or an author-supplied copy), `no_lawful_public_text_located` (only closed locations found after the documented route checks) or `route_check_incomplete`. A paper whose allowed material cannot be lawfully retained before the pilot cannot be a pilot case and is recorded as prospectively unavailable to the pilot with the routes checked; this is not a G3 input verdict, not a G1 negative and does not remove the paper from the main frame. Machine screening of retained text yields provisional G1/G2 proposals only; human adjudication against the retained bytes is required before any gate is recorded. If the investigator delegates the first pass, the delegated gates are recorded as a named primary-assessment record with a deviation entry, every gate citing segment IDs of the hash-checked retained bytes, and remain provisional: the funnel stays unassessed and no pilot selection is frozen until the investigator team verifies or overturns each gate. Delegated or human adjudication may skip a candidate only for absent lawfully retained text or for G1/G2 ineligibility, never for G3, G4, G5 or expected reconstruction difficulty, and every skipped candidate is retained with its reason in the deterministic chain.

**Amendment AMEND-2026-09-24-03** (`AMENDMENT_AMEND-2026-09-24-03_deposited_data_criterion.md`, hash-frozen and RFC 3161 timestamped before any screening under it) restricts the main reconstruction sample to papers whose deposited analysis data a third party can obtain lawfully and immediately. Under the amendment, author-request, application-required, institutional-licence, size-unannounced, above-cap, listing-unavailable and output-only deposits are sampling-frame exclusions recorded with their class, not G3 study outcomes. Papers that pass G1/G2 but fail the amended access criterion form the `ACCESSIBILITY_GATE_FAILED` layer (`data/adjudication/accessibility_gate_failed_*.json`): an audit sample of pre-reproducibility access barriers for an auxiliary analysis or a separate paper, from which no reconstruction success or failure rate may be computed because no member reached a reconstruction attempt. The seven papers of the provisional pilot `DEV-2026-09-24-02` are the first members of that layer and are not in the commercial-software reproducibility sample.

## Prospective pilot and freeze sequence

After harness qualification, select seven pilot papers, one from each nonempty disjoint field assignment, by the deterministic hash procedure in SAP. These are feasibility cases excluded from the main sample. Use main-frame replacements only if prospectively ineligible, recording every reason; do not replace difficult reconstruction outcomes.

Pilot only the same information-access intervention intended for the main study. Use it to finalize target comparability, tolerances, scoring, access procedures, budgets, stopping rules and telemetry. No post-pilot numerical choices are final until all named protocol documents, prompts, manifests, image and tool registry are hashed and externally timestamped. A Git commit or this draft's date alone does not establish prospective freeze.

Freeze all documents in this directory, resolve every `NOT_COMPLETED` gate, record the model/version available to the investigator, preregister when an appropriate public record is available, then sample the main study and seal paper-specific targets before runs. Later changes require an explicit deviation record, not a rewritten historical protocol.

## Measurement and governance

Three fixed independent slots per paper use identical prompts, packages and permissions and equivalent model/resource settings. No shared mutable state, parent debugging, sibling results or original implementation is permitted. Outcomes L1–L5 remain separate. Numerical agreement is primary; conclusion preservation is secondary and must not rescue a numerical mismatch.

Validated policy barriers remain in the all-eligible policy estimand but are not invented failed runs. Unresolved eligibility, access, contamination or missing runs remain unresolved. Agent failure cannot establish universal human impossibility or publication inadequacy.

Human scientific interpretation and final author responsibility are required. Author list, affiliations, contributions, funding, interests, ethics determination for human performance data, and APC arrangements are not supplied or presumed.

# Quality-control protocol

**DRAFT — NOT FROZEN**

First perform a reviewer-perspective critique of question/novelty, statistical design, figures/tables, reproducibility and claim strength. Rank each issue by scientific severity, expected benefit and feasibility. Mechanical formatting cannot overrule a fatal scientific gap.

Required final audits:

| Audit | Evidence required |
|---|---|
| Fabrication | Every empirical row from real source or observed run; empty human/agent tables not scored. |
| Numerical | Manuscript/table/figure values derived from canonical results and raw source hashes. |
| Denominator | Unique paper versus source row; at-risk versus full frame; missing and contaminated slots; sample weights. |
| Figures/tables | Sequential citations, no orphan elements, English labels, editable outputs and journal-appropriate placement. |
| Methods/code | Every claimed method implemented and exercised; no unperformed freeze, interval or validation claims. |
| Endpoint | Target and tolerance sealed before run; no post-hoc easiest target. |
| Firewall | Qualified broker and control-plane restriction, access telemetry, canary failures, contamination adjudication. |
| Independence | Separate contexts and mutable storage; no parent/sibling debugging. |
| Commercial policy | Software invoices/entitlements or explicit zero-purchase evidence; compute costs distinct. |
| Failure attribution | Observations and causal explanations distinguished; uncertain categories preserved. |
| Human blinding | Genuine participation, ethics/consent determination, masked packages and scores. |
| Reveal | Frozen outcomes precede original-code access; no perturbation or revised score. |
| References/novelty | Primary source, version, DOI, status and supported claim; no unsupported “first”. |
| Discussion/abstract | No inference from agent failure to human impossibility; no unavailable results. |
| Supplement | All protocol versions, deviations, schemas, rights/access paths and reproducible code. |
| Rebuild | Clean public checkout and stated inputs regenerate every submitted result without private files. |

Automated checks currently cover immutable sources, duplicate conflicts, identity/membership preservation, unknown-aware slot logic, implementation/ execution/numerical consistency, timestamp ordering, scalar rounding, sampling invariance and reference interval calculations.

These are software checks, not validation of source labels or the unimplemented experimental harness.

Before submission, repeat targeted literature and live journal-guideline checks, generate the final empirical manuscript from the frozen study data, review all author declarations, render every document and manually inspect it, and archive a versioned lawful release. `submission-check` must remain blocked until those requirements are implemented and evidenced; never weaken it to make a build green.

## Blind freeze and descriptive reveal

**FROZEN 2026-09-25 by AMEND-2026-09-25-05 (hash `161062fa…9c12` in `data/adjudication/protocol_freeze_20260925.json`). The stale `DRAFT — NOT FROZEN` header was replaced by AMEND-2026-09-25-07 after the main blind seal; no rule text changed. Applied to the main study in `data/adjudication/main-study-20260925/reveal/reveal_ledger.json`.**

Close all main blind slots, adjudicate and freeze outcomes/codes/comparisons, then hash code, inputs, outputs, public action logs, access events and manifests. Record byte counts, UTC freeze, independent timestamp receipt and artifact IDs in `blind_freeze_manifest.csv`. Seal human blind packages before reveal.

The required ordering is $\textit{target selected} < \textit{run start} < \textit{run stop} \leq \textit{blind freeze}$ < original-code reveal. No missing timestamp may be invented or backdated. A checksum alone proves content identity, not when the content existed.

A separate reveal reviewer may then inspect original code where lawfully available. Compare publication, independent implementation and original choices for normalization, filters, missing-data handling, initialization, seeds, stopping, defaults, preprocessing order, exclusions, manual transformations, constants, boundaries, data cleaning and versions. Use explicit/partial/ambiguous/not_reported for description completeness. Preserve source evidence.

Reveal is descriptive only. Do not rerun with original choices, substitute original code, perturb any discrepancy or revise the frozen blind score. Unavailable original code yields a missing reveal assessment, not a negative description-quality finding. Human validators must never see reveal material.

# Statistical analysis plan

**FROZEN 2026-09-25 by AMEND-2026-09-25-05 before main-cohort selection. No main-study outcome existed at the freeze; no pilot value had been adjudicated. Hash and RFC 3161 timestamp: `data/adjudication/protocol_freeze_20260925.json`.**

## Estimands and denominators

The population is the author-approved unique-PMID frame derived from the fixed deposit, not all publications. Its exact count, ordered CSV hash and source/decision provenance are in `results/frame_manifest.json`. Preserve source rows and all recorded field memberships in the separate bridge; no replacement publications are added. Let E denote G1=yes papers and U denote G1=uncertain papers. The proposed primary estimand is end-to-end policy demonstration among E, with all verified access barriers retained. For an assessable attempted paper the endpoint is at least two valid successes among three fixed slots. A verified indispensable gate barrier is policy non-success, with `not_attempted_gate_barrier`; it does not produce three failed runs. Unresolved gates yield an unknown endpoint. The conditional rate among attempted papers is secondary, with its own denominator. G5 insufficiency alone must not select difficult but attemptable papers out.

For an assessed gate, report yes, no and unknown/unassessed counts over both the original paper set and the actual conditional risk set. Supply identification bounds [yes/N, (yes+unresolved)/N] and the resolved-case rate separately. Do not multiply marginal gate rates to construct a funnel. Full-frame census proportions have no paper-sampling uncertainty; any binomial interval must be labelled a model-based reference, not a census sampling confidence interval. Classification uncertainty and validation sampling uncertainty require separate reporting.

## Sampling

Use disjoint assignment before looking at outcomes: among each paper's recorded field labels choose the label with minimum SHA256 of `paper2-field-v1|paper_id|field`. Retain every original membership in the bridge. After removing pilot cases and deciding eligibility, allocate about 100 main papers across nonempty assigned fields. Set one place per nonempty field, then assign each remaining place to the field with largest proportional allocation deficit, respecting its capacity; resolve ties lexicographically as implemented. Within field rank by SHA256 of `paper2-main-v1|paper_id`. Selection is reproducible and has equal within-field probability under the prespecified pseudorandom ranking model. Save seed, ordered frame hash, $N_h$, $n_h$, $\pi_h = \frac{n_h}{N_h}$ and weight $\frac{N_h}{n_h}$. The pilot cases removed are the seven `PROSPECTIVE_PILOT` papers of `PILOT-FREEZE-2026-09-25`; the seven `ACCESSIBILITY_GATE_FAILED` papers and every paper with a machine-readable exclusion class or with text inspected during any earlier candidate review are not eligible. Eligibility is the exhaustive Stage A/Stage B route record under AMEND-2026-09-24-03. Main selection is executed exactly once, after this freeze, by `src/paper2/main_selection.py`, and its record is timestamped before any selected paper's target freeze.

This field-stratified design is the conservative default. Do not add commercial, code or data cross-strata unless the pilot shows adequate occupancy and the amended allocation is frozen before outcomes. Software categories are commercial_only/open_only/both/neither_detected/unknown; a negative regex match does not establish absence. Known policy barriers can be tallied by census; sample the attemptable remainder without removing it from the headline denominator.

## Precision and estimates

The build calculates Wilson intervals for planned n=100 at assumed p of 0.20,

0.30, 0.50, 0.70 and 0.80. These are analytic unweighted binomial reference calculations, not study results or guarantees for the stratified design.

For the attemptable stratum populations, estimate the success total as $\hat{S}=\sum_{h} N_h \hat{y}$. The all-eligible policy estimate is $\frac{\hat{S}}{|E|}$, retaining verified indispensable barriers as policy non-success in $|E|$. The conditional attemptable-domain estimate is $\frac{\hat{S}}{\sum_{h} N_h}$ and must not be labelled the headline estimate.

Eligible papers with unresolved gates remain in $|E|$ and add zero/their count to lower/upper success-total bounds. For finite-population uncertainty with small strata or boundary successes, invert the hypergeometric distribution for the possible number of successful papers $K_h$ in each stratum. Use two tail probabilities $\frac{\alpha}{2H}$ for H sampled strata and $\alpha=0.05$; aggregate simultaneous lower and upper $K_h$ bounds, then divide by

| | |
|---|---|
| E | for the headline or $\sum_{h} N_h$ for the conditional domain.<br>Add unresolved-gate |

counts to the upper headline bound. Label this as a sampling-and-identification envelope when unknown gates remain. This conservative interval avoids zero estimated variance in all-success/all-failure strata. Census strata have exact totals. The estimator and interval are implemented in `src/paper2/estimation.py`. Exhaustive enumeration checks finite-population coverage for all sample sizes and success totals in populations up to 12; boundary, census, unequal-weight, unknown-outcome and denominator checks are also included. These software tests are not empirical study estimates; protocol freeze remains pending.

If a paper outcome is unresolved, calculate endpoint lower/upper estimates by assigning unresolved outcomes 0/1 solely for identification bounds, explicitly labelled. Obtain sampling bounds for each endpoint; do not present the resulting envelope as an ordinary missing-at-random confidence interval. In E plus U, worst-case eligibility bounds are $\left[\frac{L_E}{|E|+|U|}, \frac{U_E+|U|}{|E|+|U|}\right]$, where $L_E$ and $U_E$ are eligible-set lower/upper success totals. For sampled outcomes, carry sampling bounds into these limits.

## Run and paper summaries

Each valid run requires implementation, execution and target agreement. For fixed slots, k = clean successes and m = missing/invalid slots. Threshold t is achieved if $k \geq t$, failed if $k+m<t$, otherwise unknown. Use $t=2$ primary, $t=3$ strict and $t=1$ permissive. Show 0/3–3/3 only for fully assessable clean triples, and display the incomplete triple count alongside. Never reduce a contaminated triple to a new denominator.

Report each L1–L5 separately, including applicable conditional and unconditional denominators. Preserve distinct point/interval/p-value/conclusion comparisons. Failure frequency uses papers and runs as separate units; multiple codes may sum above the denominator. Burden uses median, quartiles, range and missingness, with durations of budget-censored runs identified rather than

treated as fast completed successes. Report software purchase, agent/API and compute costs separately.

## Models and sensitivities

Do not treat three runs as three independent papers. Descriptive estimates are primary. A paper-random-intercept logistic model is optional only if at least 20 distinct papers with successes and 20 with failures support every candidate predictor degree of freedom, clustering is identifiable, and no separation or unstable fit is present. Freeze any reduced predictor set before outcome access; never screen predictors by observed p values. Given the planned size and seven fields, an adjusted multi-predictor model is likely unjustified and may be omitted. A code-sharing association is observational, not causal; original code stays hidden.

Prespecified sensitivity descriptions: strict/majority/permissive; clean versus all-assigned accounting without reclassifying contaminated runs as valid; uncertain-target exclusion; lower/upper unknown bounds; field, software category and code-detection subgroups. Report sparsity; do not fish for interactions. No alternative models/defaults/implementations, perturbations, multiverse, robustness scores or Paper III experiments are permitted.

## Target selection

**FROZEN 2026-09-25 by AMEND-2026-09-25-05 (hash `fc879735acc5698093d1476db7e4d87ac4174807f5cbea0eb2068e91162a99da`); stale draft header corrected by AMEND-2026-09-25-06 without any rule change.**

Use, in order: primary numerical endpoint; principal effect; principal performance metric; central quantitative table; central quantitative figure; principal computational conclusion. Choose the paper's stated priority, then order of appearance for equally central candidates. Record alternatives considered and rejection reasons. A convenient or easy endpoint is never a replacement after seeing reconstruction output.

Require a computational, central and objectively assessable output, its exact publication/supplement location, reporting precision, input identity, estimand and prospective comparison rule. Record the minimal target-linked claim separately. If a conclusion has no defensible numerical operationalization, G2 is uncertain or no; do not silently replace the primary endpoint with a subjective judgement.

The manifest records target_id, paper_id, source_id, page/table/figure/cell, quoted claim, numerical value, units, metric, evaluation population/split, selection rationale, tolerance, timestamp and selector. Targets and packages must be hashed and sealed before any reconstruction run. The evaluator's score specification is fixed; no author implementation is used to choose a target. Same target and material are delivered to every slot. Do not reveal another agent's interpretation or an author's implementation in a clarification.

## Target-specific agreement

**FROZEN 2026-09-25 by AMEND-2026-09-25-05. Generic scalar/count comparison is implemented; a non-scalar target may only be frozen for a main-study paper if its comparison rule below is complete at that paper's prospective freeze.**

No universal percentage tolerance. Store both target and observed values, units, absolute error and meaningful relative error; relative error near zero is not informative. Review reporting precision rather than demanding hidden digits.

| Target | Prospective rule |
|---|---|
| Counts | Exact, unless publication explicitly defines stochastic/approximate counts. |
| Deterministic scalar/effect | Agreement within the displayed rounding bin. Default nearest rounding uses $\left[v-0.5\,10^{-d}, v+0.5\,10^{-d}\right)$; if truncation or another convention is explicit, use it. Declare ambiguous boundary cases prospectively. |
| Interval | Point estimate, lower bound and upper bound separately; all required components meet their individual precision rules. Inferential interpretation remains separate. |
| P value | Preserve exact reported precision or reported inequality. Significance category alone does not meet an exact-value target. |
| Proportion | Numerator and denominator when reported; otherwise displayed precision and units. |
| Prediction/performance | Identical prespecified metric, evaluation population and split. Unknown split identity is not cured by a similar score on different data. |
| Table | Prospectively identified central cells; all must meet their cell rules. Retain element errors. |
| Figure | Prefer underlying values. If extraction is unavoidable, freeze coordinates, extraction uncertainty and comparison before the run; no default pixel similarity. |
| Stochastic estimand | Compare the reported estimand/distributional summary under a prespecified criterion. Pilot fixes simulation count and error control within the resource ceiling. No optional rerunning until agreement. |
| Optimization | Objective value and constraints first. Exact decision variables only for a unique solution; equivalent optima are not failures. |

At pilot, reject an unscoreable target and proceed down the hierarchy before main runs; log the reason. No stochastic, figure or non-scalar target may be launched with an unresolved comparison rule. Numerical agreement does not itself prove faithful algorithm implementation. Review fidelity using the published method without introducing alternative analyses.

Value provenance (AMEND-2026-09-25-05 §4): an observed value counts toward L3 and L4 only if the adjudicator locates it, to its reported precision, in a journaled execution output produced by journaled code that reads served inputs. A value present only in the report is `not_established`.

DRAFT — NOT SUBMISSION READY: main study sealed; delegated mechanical adjudication; A/B/C/E AI-assisted review imported as a provisional sensitivity layer; investigator sign-off, human adjudication and human validation (Section D) pending